Lim, Hyeon Jun

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Research Interests

- My goal is to develop scalable reinforcement learning methods for complex tasks while reducing the data and effort needed to learn effective policies. I am interested in using simulation and learned world models to generate useful experience for policy learning, and in improving how policies reuse available data through off-policy learning.

Education

- **Yonsei University** — Seoul, South Korea
  M.S. in Electrical and Electronic Engineering — *Feb. 2018*
  - Thesis: "State Estimation Using the Sampled-data Fuzzy Observer and Its Application to the AHRS"
  - Brain Korea 21 Plus Scholarship, National Research Foundation of Korea (NRF)
- **Yonsei University** — Seoul, South Korea
  B.S. in Electrical and Electronic Engineering — *Feb. 2016*
  - National Science and Engineering Scholarship, Government of South Korea

Employment

- **Hyundai Motor Company** — Seoul, South Korea
  Senior Research Engineer, Autonomous Driving Development — *Jan. 2025 – Present*
- **Hyundai Motor Company** — Seoul, South Korea
  Research Engineer, Autonomous Driving Development — *Mar. 2022 – Dec. 2024*
- **Korea Institute of Industrial Technology (KITECH)** — Ansan, South Korea
  Researcher, Robotics R&BD Group — *Jun. 2018 – Jun. 2021*

Key Projects

- **Unified Closed-Loop Simulation Framework for Reinforcement Learning-Based Autonomous Driving** — *Aug. 2025 – Present*
  - Building a closed-loop RL training framework that supports multiple environment representations (3DGS scenes, graphics-based simulation, and object-level abstractions).
  - Developing scalable RL algorithms across synthetic, reconstructed, and real-world driving data.
- **End-to-End Autonomous Planning System** — *Jul. 2024 – Mar. 2026*
  - Co-developed a transformer-based end-to-end planner and built its training and inference framework.
  - Led its embedded deployment and validated it on a real vehicle.
- **Lv.3 / Lv.4 Autonomous Driving Perception Systems (HDP & Robo-Services)** — *Mar. 2022 – Jun. 2024*
  - Designed the software architecture of the LiDAR perception stack for Lv.3 and Lv.4 platforms, spanning clustering, localization, and online calibration.
  - Developed the point cloud pre-processing pipeline, CUDA-optimized for the embedded target.
- **Standardized Open Database and Evaluation System for Intelligent Vehicle Perception** — *Jun. 2018 – May 2020*
  - Built the Korea Dataset for Autonomous Solution (KODAS), the first large-scale autonomous driving dataset in Korea, with over a million labeled objects captured by 128-channel LiDAR.
  - Developed cm-level HD mapping, the point cloud processing toolchain it relied on, and multi-sensor calibration.
- **River Management Technology Using a Drone-Based Survey Platform** — *Jun. 2019 – Jun. 2021*
  - Developed a LiDAR simulation framework to derive hardware specifications and validate dual-wavelength bathymetric sensing for river and subsurface mapping.

- **3D Vehicle Shape Generation System for Automated Car Wash** *Jan. 2019 – Dec. 2020*
  - Developed a multi-LiDAR vehicle-profiling system with sub-2 cm accuracy for unmanned car-wash automation, running in real time on TI embedded hardware.
- **Vision-IMU Integrated Indoor Navigation System for the Visually Impaired** *Oct. 2016 – Sep. 2017*
  - Developed an indoor positioning system combining marker-based visual recognition with smartphone IMU dead-reckoning.

Publications

Journal Articles

- Younghun Byeon, Eunju Kim, **Hyeon Jun Lim**, et al., "Development of a Faster R-CNN-based Marine Debris Detection Model for an Embedded System," Journal of Institute of Control, Robotics and Systems, 27(12), 1038–1043 (KCI), 2021.
- Younghun Byeon, Minjae Kang, **Hyeon Jun Lim**, et al., "Development of a Three-Dimensional Indoor Position Estimation Algorithm for a Drone Using Stereo Vision and YOLOv3," Journal of Institute of Control, Robotics and Systems, 27(11), 845–850 (KCI), 2021.
- **Hyeon Jun Lim**, Young Hoon Joo, Jin Bae Park, "A Relaxed Stabilization Condition for Discrete T-S Fuzzy Model under Imperfect Premise Matching," Journal of Korean Institute of Intelligent Systems, 27(1), 59–64 (KCI, IF 0.47), 2017.

International Conferences

- **Hyeon Jun Lim**, Jin Bae Park, Young Hoon Joo, "Fuzzy Observer Design and Its Experimental Validation Using the Fuzzy Lyapunov Function," 17th International Conference on Control, Automation and Systems (ICCAS), Jeju, Korea, 2017.
- **Hyeon Jun Lim**, Jin Bae Park, Young Hoon Joo, "Fuzzy Lyapunov Function-based H∞ Fuzzy Filter Design for Sampled-data Systems under Imperfect Premise Matching," SICE Annual Conference, Kanazawa, Japan, 2017.
- **Hyeon Jun Lim**, Jin Bae Park, Young Hoon Joo, "The Stabilization Condition of Continuous Affine Fuzzy Systems Under Imperfect Premise Matching," 16th International Conference on Control, Automation and Systems (ICCAS), Gyeongju, Korea, 2016.
- **Hyeon Jun Lim**, Jin Bae Park, Young Hoon Joo, "The Fuzzy Lyapunov Function-based Relaxed Stabilization of Output-feedback Fuzzy Control Systems Under Imperfect Premise Matching," International Conference on Fuzzy Theory and Its Applications (iFuzzy), Taichung, Taiwan, 2016.

Domestic Conferences

- **Hyeon Jun Lim**, Seungho Baeg, "An Improvement Method for Global Mapping of Point Cloud Data Through the Statistical Outlier Removal," 16th Korea Robotics Society Annual Conference (KRoC), Pyeongchang, Korea, 2021.
- **Hyeon Jun Lim**, Seungho Baeg, "A Time Compensation Method for a Scanning LiDAR Sensor via Hermite Interpolation," KIIS Autumn Conference, 30(2), 2020.
- **Hyeon Jun Lim**, Seungho Baeg, "A Calibration Method for a 2D LiDAR Sensor and a Linear Encoder System Using Dihedral Angle," KIEE Conference on Information and Control Systems (CICS), Jeju, Korea, 2020.
- **Hyeon Jun Lim**, Seungho Baeg, Jae-Hak Kim, et al., "Analysis of Design Parameters and Performance Simulation for a Bathymetric LiDAR Sensor," KSCE Convention, Drone Platform-based River Survey Special Session, 2020.
- Seungho Baeg, Beomjin Hong, Suhyun Na, **Hyeon Jun Lim**, "Korea Dataset for Autonomous Solution and the Acquisition System," KSAE Autumn Conference, 2018.
- **Hyeon Jun Lim**, Jin Bae Park, Young Hoon Joo, "Fuzzy Output-feedback Controller Design for T-S Fuzzy Models based on Improved Fuzzy Lyapunov Functions under Imperfect Premise Matching," ICROS Annual Conference, Sokcho, Korea, 2017.
- **Hyeon Jun Lim**, Jin Bae Park, Young Hoon Joo, "Output-feedback Controller Design for T-S Fuzzy Models based on Fuzzy Lyapunov Functions: Guaranteed-cost Approach," KIIS Spring Conference, 27(1), 2017.

- **Hyeon Jun Lim**, Jin Bae Park, Taek Gyu Song, et al., "The Relaxed Stabilization Condition for the Affine T-S Fuzzy Systems via Improved Fuzzy Lyapunov Functions: Guaranteed Cost Approach," KIEE Summer Conference, 48th, 2017.
- **Hyeon Jun Lim**, Chang-Gil Lee, Jin Bae Park, et al., "Fuzzy Output-feedback Control Design for T-S Fuzzy Models under Imperfect Premise Matching," KIEE Conference on Information and Control Systems (CICS), 2016.
- **Hyeon Jun Lim**, Han Sol Kim, Jin Bae Park, et al., "Stabilization for the T-S Fuzzy Systems Using the Line-integral Fuzzy Lyapunov Function," KIIS Spring Conference, 26(1), 2016.
- **Hyeon Jun Lim**, Young Hoon Joo, Jin Bae Park, "A Relaxed Stabilization Condition for Discrete T-S Fuzzy Model under Imperfect Premise Matching," KIIS Autumn Conference (Best Paper Award; later extended into the 2017 JKIIS article), 2016.

Patents

- "Apparatus and Method for Processing Trajectory Data," KR 10-2026-0139322 (Application), Filed Jul 28, 2026.
- "System For Processing Data Required For Object Detection Based On Artificial Intelligence and Multiple Autonomous Driving Sensors and Method Implementing The Same," KR 10-2025-0090242 (Application), Filed Jul 4, 2025. Also filed: US 19/415,494 · CN 202511840140.4.
- "Apparatus For Controlling Vehicle and Method Thereof," KR 10-2025-0044855 (Application), Filed Apr 7, 2025. Also filed: US 19/391,095 · DE 10 2025 146 420.0 · CN 202511797806.2.
- "System and Method For Calibrating 3D Sensor Coordinate Based On Scene Flow Network," KR 10-2025-0042966 (Application), Filed Apr 2, 2025. Also filed: US 19/406,040 · CN 202511849897.X.
- "System and Method For Adaptive Calibration For Autonomous Driving," KR 10-2025-0042965 (Application), Filed Apr 2, 2025. Also filed: US 19/409,994 · CN 202511849891.2.
- "3D Vehicle Shape Data Acquisition Method and System for Automated Car Wash Equipment," KR 10-2504548 (Registered), Filed Dec 7, 2020 · Registered Feb 23, 2023.
- "3D Vehicle Shape Data Acquisition Method and System for Automated Car Wash Equipment, Rotating LiDAR Configuration," KR 10-2504547 (Registered), Filed Dec 7, 2020 · Registered Feb 23, 2023.

Software Copyrights

- "Laser Beam Characteristics Evaluation Program," Korea Copyright Commission, C-2021-008734, Feb 2021.
- "LiDAR Sensor Data Viewer Program," Korea Copyright Commission, C-2020-052146, Dec 2020.
- "LAS File Conversion Program," Korea Copyright Commission, C-2020-052137, Dec 2020.
- "NovAtel GNSS Data Reception Firmware for TI ARM Processor," Korea Copyright Commission, C-2020-052144, Dec 2020.
- "Range Data Acquisition Firmware for DSP Processor," Korea Copyright Commission, C-2020-052143, Dec 2020.
- "PRU Firmware for Motor Speed Control and Rotation Angle Detection on TI DSP," Korea Copyright Commission, C-2020-052142, Dec 2020.
- "CAN and Serial Communication API Driver for TI DSP," Korea Copyright Commission, C-2020-052141, Dec 2020.
- "Epson IMU Data Reception Firmware for PRU Processor," Korea Copyright Commission, C-2020-052140, Dec 2020.
- "Drone-mounted Laser Beam Parameter Simulator," Korea Copyright Commission, C-2019-038711, Dec 2019.

Invited Talks

- "State Estimation Using the Sampled-data Fuzzy Observer and Its Application to the AHRS," Invited talk, Korea Maritime and Ocean University *Feb 2020*

Awards and Honors

- **Best Paper Award, KIIS Autumn Conference**, Korean Institute of Intelligent Systems *2016*
- **Brain Korea 21 Plus Scholarship**, National Research Foundation of Korea (NRF) *2016 - 2018*
- **National Science and Engineering Scholarship**, Government of South Korea *2012 - 2016*

Teaching

- Technical Instructor — Real-time LiDAR-to-Vehicle Calibration and Evaluation Methodologies, Hyundai Motor Group *Nov 2023 – May 2024*
- Technical Instructor — Research and Implementation of LiDAR Calibration Technology, Hyundai Motor Group *May 2023 – Oct 2023*
- Teaching Assistant — Control Engineering, Yonsei University *Fall 2016*
- Teaching Assistant — EE Experiments: Fundamentals, Yonsei University *Spring 2016*
- Teaching Assistant — Data Structures and Algorithms, Yonsei University *Spring 2016*

Academic Service

- Reviewer, Journal of Electrical Engineering & Technology (JEET) *Dec 2025*
- Reviewer, Journal of Korean Institute of Intelligent Systems (KIIS) *Mar 2024*
- Reviewer, Journal of Institute of Control, Robotics and Systems (ICROS) *Aug 2023*

Technical Skills

- **Languages** — Python, C++, MATLAB
- **Deep Learning & AI** — PyTorch, TensorFlow
- **Robotics & Perception** — ROS, PCL, OpenCV
- **Control & Simulation** — MATLAB/Simulink, NVIDIA AlpaSim, AlpaGym, Applied Intuition
- **Embedded Platforms** — NVIDIA, Qualcomm, TI
- **Sensors** — 128-ch LiDAR, 2D/3D Scanning LiDAR, Radar, GNSS/IMU, Multi-view / AVM Camera